

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Winter Examination – 2024**

**Course: B.Tech Branch : Mechanical Engg/Mechanical Engineering(Sandwich)**

**Semester :VI**

**Subject Code & Name: BTMEC604B\_Y19 IC Engines**

**Max Marks: 60**

**Date:26/12/2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                                                                                                                                                                                  | (Level/CO) | Marks     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                                                                                                                                                                           |            | <b>12</b> |
| 1           | The air standard efficiency of an I.C. engine depends on<br>a. fuel used      b. speed of engine      c. compression ratio      d. none of the mentioned                                                                                                                         | <b>CO1</b> | 1         |
| 2           | The Otto cycle consists of _____<br>a. two constant pressure processes and two constant volume processes      b. two constant pressure and two constant entropy processes      c. two constant volume processes and two constant entropy processes      d. none of the mentioned | <b>CO1</b> | 1         |
| 3           | For same compression ratio and for same heat added _____<br>a. Otto cycle is more efficient than Diesel Cycle      b. Diesel cycle is more efficient than Otto Cycle      c. Efficiency depends on other factors      d. None of the mentioned                                   | <b>CO1</b> | 1         |
| 4           | The knock in diesel engine occurs due to _____<br>a. instantaneous and rapid burning of the first part of the charge      b. instantaneous auto ignition of last part of charge      c. reduction of delay period      d. none of the mentioned                                  | <b>CO1</b> | 1         |
| 5           | Which is the most popular firing order in the case of four cylinder in-line I.C. engine?<br>a. 1-2-3-4      b. 1-3-2-4      c. 1-4-2-3      d. 1-3-4-2                                                                                                                           | <b>CO1</b> | 1         |

|      |                                                                                                                                                    |                                               |                                              |                          |      |    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|--------------------------|------|----|
| 6    | The total power developed by combustion of fuel in the combustion chamber is called _____                                                          |                                               |                                              |                          | CO1  | 1  |
|      | a. brake power                                                                                                                                     | b. indicated power                            | c. mean effective pressure                   | d. none of the mentioned |      |    |
| 7    | The frictional power (F.P) is given by _____                                                                                                       |                                               |                                              |                          | CO1  | 1  |
|      | a. $F.P. = B.P. - I.P.$                                                                                                                            | b. $F.P. = I.P. - B.P.$                       | c. $F.P. = B.P. / I.P.$                      | d. $F.P. = I.P. / B.P.$  |      |    |
| 8    | Paraffins have molecular structure of _____                                                                                                        |                                               |                                              |                          | CO1  | 1  |
|      | a. chain saturated                                                                                                                                 | b. chain unsaturated                          | c. ring saturated                            | d. ring unsaturated      |      |    |
| 9    | The function of a float chamber in a carburettor is to supply the fuel to the nozzle at a _____                                                    |                                               |                                              |                          | CO1  | 1  |
|      | a. constant volume head                                                                                                                            | b. constant pressure head                     | c. variable pressure head                    | d. none of the mentioned |      |    |
| 10   | The injection of fuel into a SI engine can be done by _____                                                                                        |                                               |                                              |                          | CO1  | 1  |
|      | a. direct injection of fuel into the cylinder                                                                                                      | b. injection of fuel close to the inlet valve | c. injection of fuel into the inlet manifold | d. all of the mentioned  |      |    |
| 11   | In magneto ignition system, as the engine speed increases the flow of current _____                                                                |                                               |                                              |                          | CO1  | 1  |
|      | a. decreases                                                                                                                                       | b. remains constant                           | c. increases                                 | d. none of the mentioned |      |    |
| 12   | Which is the most preferred S.I. engines fuel?                                                                                                     |                                               |                                              |                          | CO1  | 1  |
|      | a. aromatics                                                                                                                                       | b. paraffins                                  | c. olefins                                   | d. naphthenes            |      |    |
|      |                                                                                                                                                    |                                               |                                              |                          |      |    |
| Q. 2 | Solve the following.                                                                                                                               |                                               |                                              |                          |      | 12 |
| A)   | Explain with the help of P-θ diagram, different stages of combustion in Compression Ignition engine                                                |                                               |                                              |                          | CO-3 | 6  |
| B)   | Draw a valve timing diagram for a four stroke C.I engine and explain.                                                                              |                                               |                                              |                          | CO-4 | 6  |
|      |                                                                                                                                                    |                                               |                                              |                          |      |    |
|      |                                                                                                                                                    |                                               |                                              |                          |      |    |
| Q.3  | Solve the following.                                                                                                                               |                                               |                                              |                          |      | 12 |
| A)   | Explain in detail the phenomenon of Ignition delay with respect to Compression Ignition Engine. List down the factors affecting the ignition delay |                                               |                                              |                          | CO-4 | 6  |
| B)   | Describe the operation of splash lubrication system with the help of neat sketch                                                                   |                                               |                                              |                          | CO-3 | 6  |
|      |                                                                                                                                                    |                                               |                                              |                          |      |    |
| Q. 4 | Solve Any Two of the following.                                                                                                                    |                                               |                                              |                          |      | 12 |
| A)   | What are the requirements of an ignitions system of a spark-ignition engine? List down the different types of ignition systems?                    |                                               |                                              |                          | CO-3 | 6  |
| B)   | What are the methods used for governing of I.C. engines? Explain in detail the governing method utilized in spark-ignition engine.                 |                                               |                                              |                          | CO-4 | 6  |

|      |                                                                                                                                         |      |    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------|------|----|
| C)   | Define Supercharging of an IC Engine. What are the its objectives? what is the effect of altitude on power output of the engine         | CO-4 | 6  |
|      |                                                                                                                                         |      |    |
| Q.5  | Solve Any Two of the following.                                                                                                         |      | 12 |
| A)   | Draw the typical performance plot speed characteristics curves for a SI engine and discuss the nature of the curve                      | CO-3 | 6  |
| B)   | Write a short note on air pollution due to IC engines. List down the sources of pollutants in I.C.Engine? explain its effect on ecology | CO-3 | 6  |
| C)   | Explain alcohol as alternative fuels for IC engines bringing out their merits and demerits.                                             | CO-4 | 6  |
|      |                                                                                                                                         |      |    |
| Q. 6 | Solve Any Two of the following.                                                                                                         |      | 12 |
| A)   | Write short note on Alternative fuel for IC Engine: Biodiesel with its merits and demerits                                              | CO-3 | 6  |
| B)   | What are the types of fuel cell? Explain anyone.                                                                                        | CO-4 | 6  |
| C)   | Compare Electric vehicle with Hybrid Electric vehicle and list advantages of Hybrid vehicle                                             | CO-4 | 6  |
|      | *** End ***                                                                                                                             |      |    |